Secure Deployment in OpenStack

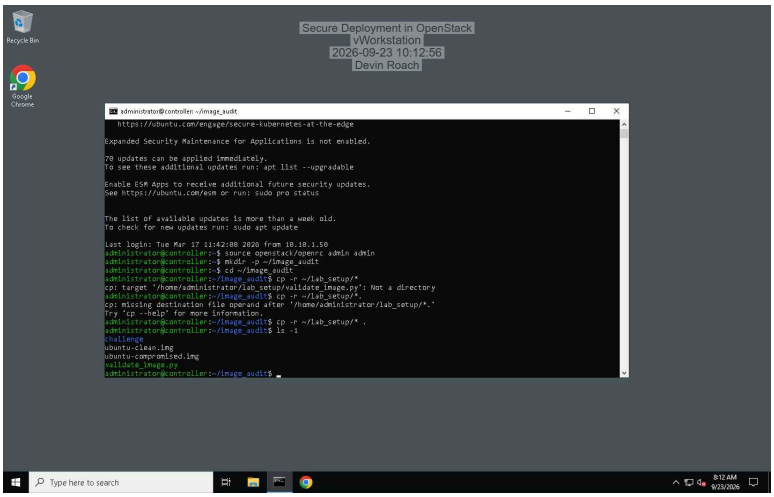
Fundamentals of Cloud Security, First Edition - Lab 04

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Time on Task: 6 hours, 56 minutes
Progress: 100%
Report Generated: Sunday, September 27, 2026 at 8:02 PM

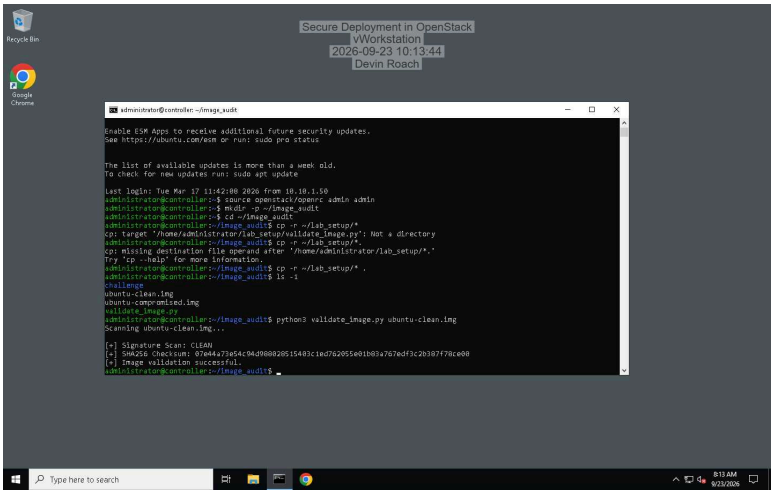
Section 1: Hands-On Demonstration

Part 1: Validate Image Integrity

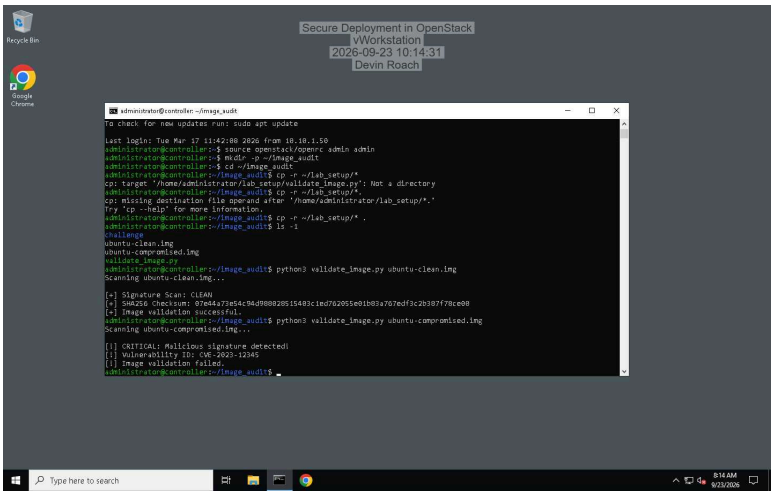
9. **Make a screen capture** of the file list showing **validate_image.py, ubuntu-clean.img, and ubuntu-compromised.img**.



11. **Make a screen capture** of the **console output showing the message "Image validation successful"**.

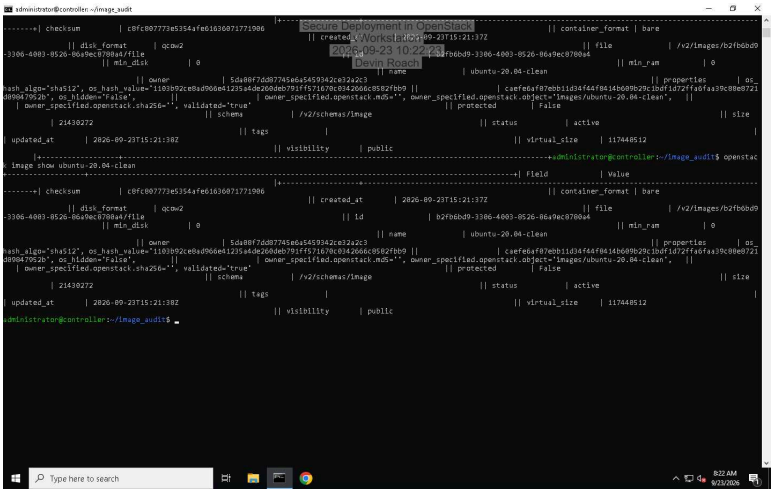


13. **Make a screen capture** of the **console output showing the message "Image validation failed" and the detected vulnerability ID**.

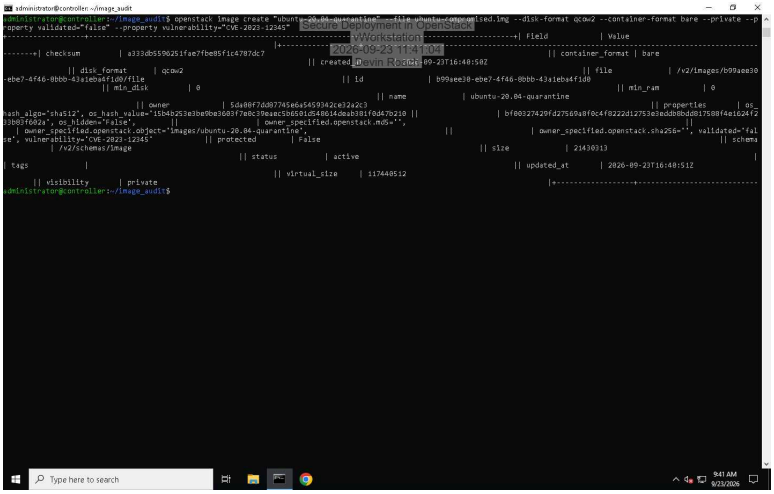


Part 2: Upload and Tag Validated Images

3. **Make a screen capture** of the **output showing the properties table with validated='true' listed**.

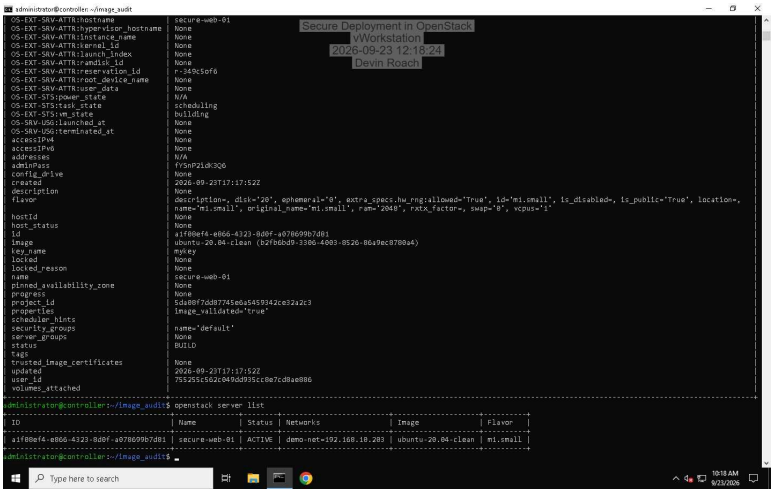


6. **Make a screen capture** of the **image list command output showing both the clean and quarantined images**.

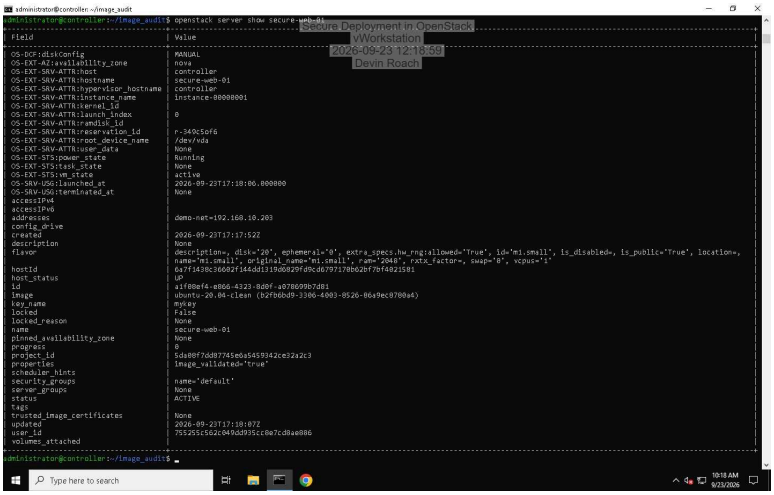


Part 3: Deploying from Trusted Assets

3. **Make a screen capture** showing the **secure-web-01 instance in the ACTIVE state**.



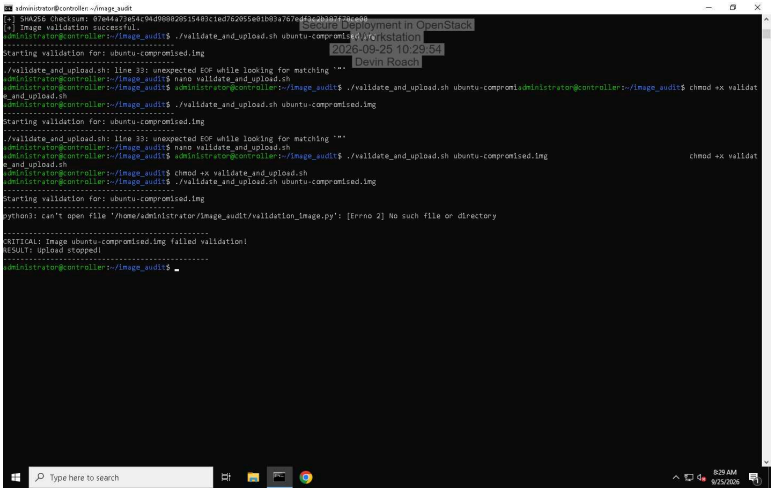
5. **Make a screen capture** of the **output, specifically highlighting the image section where the properties are listed**.



Section 2: Applied Learning

Part 1: Automate the Security Pipeline

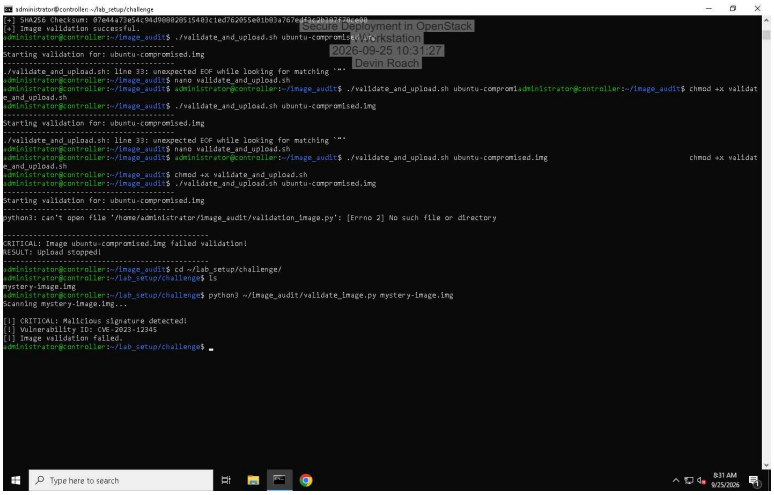
7. **Make a screen capture** of the **terminal showing your script correctly printing "CRITICAL: Image failed validation"**.



Section 3: Challenge and Analysis

Part 1: The Trojan Horse

4. **Make a screen capture** of the **result**.



6. **Submit the command** you used to upload and tag the mystery image.

#!/bin/bash
IMAGE_FILE=

6. **Submit the command** you used to upload and tag the mystery image.

IMAGE_NAME=$(basename "$IMAGE_FILE")
if [-z "

6. **Submit the command** you used to upload and tag the mystery image.

"]; then echo "Usage:

6. **Submit the command** you used to upload and tag the mystery image.

" exit 1fi
echo "----------"echo "Starting validation for: $IMAGE_FILE"echo "----------"
if python3 validate_image.py "$IMAGE_FILE"; then echo "" echo "[SUCCESS] Validation Passed. Checking OpenStack Auth..." echo "[ACTION] Uploading $IMAGE_NAME to Glance..."
openstack image create "$IMAGE_NAME" \ --file "$IMAGE_FILE" \ --disk-format qcow2 \ --container-format bare \ --public \ --property validated="true" \ --property investigation_status="active"
echo "----------" echo "Upload Complete: $IMAGE_NAME is now live." echo "Investigation Status: active" echo "----------"else echo "" echo "----------" echo "CRITICAL: Image $IMAGE_FILE failed validation!" echo "RESULT: Upload stopped!" echo "----------" exit 1fi